5 (a)	The magnetic flux density is defined as the force acting per unit current per unit length on a conductor placed perpendicular to the magnetic field	[1] [1]
(b)(i)	Protons are positively charged thus (current is in direction of the motion of the proton) and magnetic force towards the centre of the circular path (may describe direction of magnetic force at a specific point) By Fleming left hand rule, using the direction of current and force, the magnetic field must be out of the paper.	[1] [1]
(ii)	The centripetal force is provided by the magnetic force $F_B = F_C$ $Bqv = mv^2/r$ $v = Ber/m$ $\frac{1}{2} mv^2 = \frac{1}{2} m (Ber/m)^2$ $KE = \frac{B^2 e^2 r^2}{2m}$	[1] [1]
(iii)	E-field direction is in same direction to the motion of the proton whenever the proton crosses the gap. The electric force acting on the proton causes it to accelerate continuously each time it crosses the gap thus KE increases.	[1] [1]